

HIERARCHICAL qPCR MARKER PANELS FOR COARSE-TO-FINE CELL-TYPE FRACTION QUANTIFICATION

Anonymous authors

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ABSTRACT

Quantifying cell-type fractions from bulk molecular measurements is challenging when biologically related cell types share expression programs, creating a trade-off between assay compactness, specificity, and quantitative accuracy. We develop a discovery-only hierarchical marker-panel framework for coarse-to-fine cell-type fraction estimation from single-cell RNA-seq data: cluster-level expression signatures define reproducible lineage partitions, parent and conditional child marker panels are selected from discovery cells, and independently fitted parent and child fraction estimates are combined to obtain target fractions. We evaluate two-, three-, and four-lineage configurations, including a pseudobulk-size sensitivity analysis, using discovery-derived training mixtures and independent holdout-derived test mixtures from a PBMC dataset. Among the completed configurations, the two-lineage hierarchy with 100-cell pseudobulks provided the strongest hierarchical detection performance, with ROC AUC 0.9745, rare-target precision-recall AUC 0.9262, and sensitivity 0.8253 at fixed specificity, but its quantitative MAE was 0.0996. The matched flat multi-gene panel achieved macro ROC AUC 0.9816 and macro MAE 0.0640, while the best single-marker baseline achieved macro ROC AUC 0.9823 and macro MAE 0.0658. These results show that coarse-to-fine marker panels can provide strong structured detection in favorable hierarchy configurations, but parent-child error and conditional calibration remain important limitations for quantitative estimation; consequently, a flat assay may be preferable when fraction accuracy is the primary objective.

1 INTRODUCTION

Estimating the composition of heterogeneous biological samples from bulk molecular measurements is important for applications such as immune profiling, disease characterization, and assay development. In a bulk sample, the measured expression of a marker reflects both the abundance of its target cell type and contributions from other cell types, making marker selection central to reliable fraction estimation. A compact qPCR assay is particularly attractive because it can provide targeted, scalable, and comparatively accessible measurements, but compactness must be balanced against the specificity needed to distinguish related populations.

The problem is difficult for several related reasons. Cell types that are close in expression space can share broad lineage programs while differing in only a small number of discriminative genes. A marker that is useful for detecting a lineage may therefore be insufficient for separating its sibling populations, while a highly specific marker may provide weak quantitative signal when the target is rare. In addition, fraction estimation must remain accurate across low, intermediate, and high abundance regimes and against backgrounds enriched for biologically similar clusters. These constraints create a tension between a flat panel that treats every target independently and a hierarchical assay that first identifies a broad lineage and then resolves a target within that lineage. Hierarchical designs may reduce irrelevant comparisons and improve assay organization, but their estimates can also inherit error from both the parent and conditional child models.

We study a coarse-to-fine marker-panel framework designed from single-cell RNA-seq data but evaluated on bulk-like pseudobulk mixtures. The framework uses a cluster-stratified discovery/holdout split: normalized discovery-cell signatures determine a reproducible agglomerative hierarchy, and discovery-only differential-expression candidates support the selection of parent-lineage and con-

ditional child marker panels. Parent fractions and within-lineage conditional fractions are modeled separately, and the target fraction is estimated as the clipped product of the two predictions. Reference genes are selected for normalization and excluded from marker candidates, preserving a distinction between normalization and target discrimination. This construction provides a direct way to examine both the practical compactness of hierarchical panels and the error introduced when a target estimate is composed from multiple stages.

We verify the framework using the supplied PBMC single-cell dataset and independent holdout-derived pseudobulk mixtures. The evaluation compares hierarchical configurations with two, three, and four top-level lineages, and includes a pseudobulk-size sensitivity test that changes the number of cells per hierarchical mixture while retaining the discovery-derived assay design. Flat multi-gene panels and best-single-marker baselines are evaluated under the corresponding held-out protocols. We assess held-out detection and quantitative fraction estimation using ROC AUC, rare-target precision-recall AUC, sensitivity at fixed specificity, MAE, RMSE, correlation, bias, abundance-stratified error, and parent-versus-conditional error decomposition. This design tests not only whether hierarchical structure can improve detection, but also whether that benefit translates into more accurate fraction quantification.

Our contributions are:

- We formulate a discovery-only, coarse-to-fine marker-panel workflow for qPCR-style cell-type fraction estimation from single-cell expression data, including discovery-derived hierarchy construction, reference-gene normalization, and parent/child marker selection.
- We implement a hierarchical pseudobulk evaluation protocol in which parent-lineage and conditional child models are fitted independently and combined into a target-fraction estimate.
- We evaluate sensitivity to hierarchy granularity and pseudobulk size using independent holdout mixtures, while comparing against flat multi-gene and best-single-marker assays on matched protocols.
- We explicitly decompose hierarchical error into parent-lineage and conditional child components, exposing conditions in which hierarchical organization can aid detection yet remain unsuitable for quantitative estimation.

More broadly, the framework is intended as a transparent test bed for deciding when assay structure is worth the additional modeling and measurement complexity. Future extensions could incorporate joint constrained deconvolution, stronger calibration of conditional estimates, richer hierarchy-selection criteria, and uncertainty estimates that account for both cell sampling and model error.

2 RELATED WORK

Bulk transcriptomic deconvolution methods estimate cell-type proportions from aggregate expression profiles using reference information about the constituent populations. CIBERSORT uses a predefined signature matrix and a regression-based mixture model, while CIBERSORTx extends this setting to reference-based estimation and reference-free signature inference. These approaches are designed for broad transcriptomic measurements and typically estimate multiple population proportions from a shared bulk profile. Our method instead uses a compact, discovery-derived marker set and a scalar normalized panel score for each target; its hierarchy explicitly separates lineage detection from within-lineage discrimination. Consequently, CIBERSORT and CIBERSORTx are relevant conceptual alternatives, but they are not direct experimental baselines in the present study because the completed evaluation does not provide their full-transcriptome signature-matrix inputs or implement their inference procedures.

MuSiC and BisqueRNA use single-cell expression data to construct or weight reference profiles for deconvolving bulk samples. Their use of cell-level reference data is related to our discovery-cell design, but their central objective is reference-based estimation of mixture proportions from bulk transcriptomes rather than construction of a compact hierarchical assay. In particular, the present method does not fit a joint whole-transcriptome deconvolution model: it selects discovery-only parent and child panels, fits separate linear models for parent and conditional fractions, and combines their predictions as a product. MuSiC and BisqueRNA therefore provide important methodological context

but are not direct numerical comparators under the supplied experiment, which evaluates qPCR-style scores on targeted pseudobulks.

Targeted marker-panel approaches share our emphasis on assay compactness and interpretable genes, but they differ in how marker sets are organized and evaluated. A flat panel treats each target independently, whereas our hierarchical design assigns markers to a broad parent-lineage problem and a conditional sibling-level problem. The completed experiments directly compare this organization with a flat greedy multi-gene panel and a best-single-marker baseline on matched held-out pseudobulk protocols. They do not implement an additional published qPCR or targeted-panel method, so such studies cannot be treated as quantitative baselines here; they are relevant only for motivating the practical appeal of compact assays.

The comparison is therefore based on modeling assumptions rather than a claim that one formulation subsumes the others. Reference-based deconvolution methods require transcriptome-level measurements and reference signatures or profiles, while the present framework assumes a discovery/holdout split, a small measured marker set, and a parent-child factorization of the target fraction. These differences explain why the completed experiments compare hierarchical, flat-panel, and single-marker designs directly, while treating CIBERSORT/CIBERSORTx, MuSiC, BisqueRNA, and targeted-panel studies as related work rather than adding unsupported numerical results for them. The supplied bibliography currently contains no entries for these named methods, so no citations are added in this section.

3 BACKGROUND

Bulk molecular measurements aggregate signals from many cells, so the expression profile of a sample reflects both the expression programs of its constituent cell types and their relative abundances. Let $\mathcal{C}$ be the set of cell-type clusters, and let π be their fractions, where $\pi \in [0, 1]^K$ and $\sum_{k=1}^K \pi_k = 1$. For cluster c_k , let $\mu_k \in \mathbb{R}^G$ be its average expression profile over G genes. An idealized bulk profile can be written as

$$\mathbf{x} \approx \sum_{k=1}^K \pi_k \mu_k + \varepsilon,$$

where ε represents biological variation, measurement noise, and sampling error. Cell-fraction estimation is therefore a deconvolution problem: infer one or more entries of π from a comparatively small set of measured molecular features.

A marker panel is a selected subset of genes intended to provide information about a target cluster or lineage. For a target marker set M and normalization-only reference set R , we use the score

$$s_M(\mathbf{x}) = \frac{1}{|M|} \sum_{g \in M} x_g - \frac{1}{|R|} \sum_{g \in R} x_g.$$

This score is a Δ -expression proxy for a qPCR-style assay because the supplied data contain expression values rather than physical cycle-threshold measurements. The reference genes are selected for relative stability across discovery clusters and are excluded from marker candidates. A linear model maps the resulting score to a fraction estimate, which is clipped to the interval $[0, 1]$. This formulation treats normalization and target discrimination as separate design objectives while retaining a compact, interpretable assay representation.

Marker discovery and model evaluation must be separated when a single-cell dataset is used both to design and test an assay. We partition cells within each cluster into a discovery set and a holdout set. Discovery cells are used to rank differential-expression candidates, select reference genes, construct cluster signatures, choose marker panels, and fit fraction models. Holdout cells are reserved for generating independent pseudobulk test mixtures. This cluster-stratified protocol preserves representation of every cluster in both partitions while ensuring that the reported test predictions do not determine the assay design.

Pseudobulk simulation approximates a bulk sample by sampling cells from the available single-cell populations and averaging their expression profiles. For a target cluster, targeted mixtures vary the target fraction across low, medium, and high abundance regimes while distributing the remaining cells among non-target clusters. In the hierarchical setting, the parent-lineage fraction and the target's

conditional fraction within that lineage are sampled separately. If p denotes the parent fraction and q denotes the conditional target fraction, the target fraction satisfies $\pi_{\text{target}} = pq$. Training mixtures are generated from discovery cells, whereas test mixtures are generated independently from holdout cells. This construction makes rare targets, dominant targets, and backgrounds containing related clusters explicit parts of the evaluation setting.

A hierarchical assay organizes targets into a tree or a set of top-level lineages. For target c_k , let $\ell(k)$ denote its parent lineage. The parent model estimates

$$p_{\ell(k)} = \sum_{c_j \in \ell(k)} \pi_j,$$

and a conditional child model estimates

$$q_{k|\ell(k)} = \frac{\pi_k}{p_{\ell(k)}} \quad \text{when } p_{\ell(k)} > 0.$$

The final target estimate is

$$\hat{\pi}_k = \text{clip}(\hat{p}_{\ell(k)} \hat{q}_{k|\ell(k)}, 0, 1).$$

The method assumes that parent and conditional child signals can be modeled separately and that their product is a useful estimator of the target fraction. This factorization is attractive because it decomposes a multi-class discrimination problem into lineage-level and sibling-level problems, but it also makes the final estimate dependent on both component models.

The same fraction estimator supports two related but distinct objectives. For detection, a target is labeled positive when its fraction is at least the fixed threshold $\tau = 0.35$, and continuous predictions are evaluated by ranking-based metrics such as ROC AUC and average precision. For quantification, predictions are compared with continuous target fractions using absolute and squared error, correlation, bias, and abundance-stratified error. Detection performance does not imply accurate fraction calibration: a model may rank samples correctly while systematically over- or underestimating their abundances. This distinction is essential when comparing hierarchical structure with flat panels, because hierarchy may alter detection behavior without improving quantitative accuracy.

4 METHOD

Our method consists of four stages: cluster-stratified data partitioning, discovery-only marker and reference-gene selection, pseudobulk assay simulation, and held-out fraction prediction. All assay-design operations use the discovery cells, while the holdout cells are used only to generate independent test mixtures. For each target, we construct either a flat marker panel or a hierarchical parent/child panel, fit a one-dimensional linear fraction model from normalized panel scores, and clip predictions to $[0, 1]$.

4.1 DISCOVERY/HOLDOUT PARTITION AND CANDIDATE GENERATION

For every cluster, we randomly permute its cells using a fixed random seed and assign approximately 70% to discovery and the remainder to holdout. The split is performed independently within each cluster, with at least two cells retained in each partition. Discovery cells are used for differential-expression ranking, reference-gene selection, hierarchy construction, marker selection, and model training. Holdout cells are not used in any of these steps and are reserved for test-mixture generation.

We use Scanpy’s Wilcoxon rank-sum procedure to rank genes separately for each cluster on the discovery cells. For each cluster, we retain up to 30 ranked genes as its candidate list. The union of these lists, together with the selected reference genes, defines the reduced gene matrix used by the downstream simulations and models. Candidate markers are selected independently for each target from its discovery-derived ranked list.

4.2 REFERENCE NORMALIZATION AND PANEL SCORES

Reference genes are selected from genes outside the union of all discovery-derived marker candidates. For each gene, we compute its mean expression across discovery cells, its expression fraction, and the standard deviation of its cluster means. We rank genes by the ratio of between-cluster standard

deviation to absolute overall mean, retain genes expressed in at least 20% of discovery cells, and take the three most stable genes from a pool of at most 200 candidates. These genes are used for normalization only and cannot subsequently be selected as markers.

For marker panel M and reference set R , the score for a bulk expression vector $\mathbf{x}$ is

$$s_M(\mathbf{x}) = \frac{1}{|M|} \sum_{g \in M} x_g - \frac{1}{|R|} \sum_{g \in R} x_g.$$

For each target, this scalar score is the feature supplied to a linear regression model. The model is fitted on simulated training mixtures and its predictions are clipped to the unit interval. Because the input matrix contains expression values rather than cycle-threshold measurements, this difference is used as a qPCR-style Δ -expression proxy rather than interpreted as a physical qPCR quantity.

4.3 FLAT MARKER-PANEL CONSTRUCTION

For each target cluster, we generate targeted training mixtures from discovery cells and vary the target fraction across low, medium, and high abundance regimes. The candidate gene that gives the lowest five-fold cross-validated MAE for linear prediction of the target fraction is selected first. We then greedily consider adding remaining candidates, accepting an addition only when it improves cross-validated MAE by at least 0.002, until no qualifying improvement remains or the panel contains five genes. The best single marker from the same candidate pool is retained as the baseline comparator.

After panel selection, we fit a linear regression of the target fraction on the normalized panel score using all training mixtures. We apply the fitted model to independently generated holdout mixtures to obtain the flat-panel prediction. The best-single-marker baseline is fit in the same way, replacing the selected panel with its best individual marker; thus, the two flat alternatives use matched training and test mixtures.

4.4 HIERARCHICAL ASSAY CONSTRUCTION

For the hierarchical assay, we first compute one signature per cluster by averaging discovery-cell expression over the reduced gene set and subtracting the mean expression of the reference genes. We then apply Ward-linkage agglomerative clustering to these signatures and cut the resulting clustering into a configured number of top-level lineages. The experiments vary this number across two, three, and four lineages. The resulting lineage assignment is fixed before marker selection and is used to define the parent and child discrimination problems.

For a target cluster c_k in lineage $\ell(k)$, the parent problem treats all discovery cells from $\ell(k)$ as positive and cells outside that lineage as negative. Its candidate list is the deduplicated union of the ranked candidate lists of clusters in that lineage. The conditional child problem treats discovery cells from c_k as positive and the other cells in $\ell(k)$ as negative, using the target's ranked candidate list. For each problem, candidate genes are ranked by the absolute difference between the positive and negative mean expression, and up to three genes are retained for the corresponding parent or child panel. If a panel has no valid candidate, the implementation falls back to a valid candidate from the corresponding candidate list.

For each target, hierarchical training mixtures independently vary the parent-lineage fraction and the target's conditional fraction within that lineage. The parent model regresses the parent fraction on the normalized parent-panel score, while the child model regresses the conditional fraction on the normalized child-panel score. Let $\hat{p}_{\ell(k)}$ and $\hat{q}_{k|\ell(k)}$ denote their clipped predictions. The target estimate is

$$\hat{\pi}_k = \text{clip}(\hat{p}_{\ell(k)} \hat{q}_{k|\ell(k)}, 0, 1).$$

Both models are fitted using discovery-cell mixtures and evaluated on independently sampled holdout-cell mixtures. Parent, conditional, and product predictions are retained separately to quantify the contributions of the two stages.

4.5 PSEUDOBULK SIMULATION AND EVALUATION OUTPUTS

For flat evaluation, each simulated mixture contains 100 cells. A target fraction is first sampled from one of three regimes: $[0.01, 0.10]$ for low abundance, $[0.10, 0.50]$ for medium abundance, or

[0.50, 0.90] for high abundance, with probabilities 0.30, 0.45, and 0.25, respectively. The target count is rounded to the specified mixture size and constrained to leave at least one non-target cell. Remaining cells are allocated among all other clusters using a Dirichlet draw followed by multinomial sampling, and the mixture expression is the mean expression of the sampled cells.

For hierarchical evaluation, the parent fraction is sampled uniformly from [0.05, 0.90] and the conditional target fraction uniformly from [0.02, 0.95]. Their product determines the target fraction before integer cell counts are formed. The parent lineage receives the rounded parent count, the target receives the rounded product count, sibling clusters fill the remaining parent-lineage cells, and clusters outside the lineage fill the remaining cells in the mixture. Sibling and outside-lineage allocations are sampled with Dirichlet-multinomial draws. Training mixtures use discovery cells and test mixtures use holdout cells; the hierarchical pseudobulk size is varied across the completed configurations.

For flat panels, we report per-cluster and macro-averaged ROC AUC, Spearman and Pearson correlation, MAE, RMSE, R^2 , and low- and high-fraction MAE, together with the corresponding single-marker quantities. A target is labeled positive when its true fraction is at least 0.35. For hierarchical predictions, we additionally compute average precision, sensitivity at a threshold chosen from the 95th percentile of predictions on negative test mixtures, bias, parent-fraction MAE, and conditional-fraction MAE. The product-propagation diagnostic compares the final product prediction with the stored product; because the final prediction is defined by that product, this diagnostic is an implementation consistency check rather than an independent estimate of total hierarchy error.

5 EXPERIMENTAL SETUP

We use the provided PBMC single-cell RNA-seq dataset in H5AD format. The dataset contains a gene-expression matrix and Leiden cluster assignments in `adata.obs["leiden"]`; these clusters define the cell populations whose fractions are estimated. The cluster labels are converted to strings and sorted to provide a consistent ordering across marker discovery, pseudobulk construction, and evaluation. We do not impose additional biological annotations beyond the supplied cluster labels.

Cells are split independently within each Leiden cluster using random seed 42. Approximately 70% of the cells in every cluster are assigned to discovery and the remainder to holdout, with at least two cells retained in each partition. Discovery cells are used for Wilcoxon marker ranking, reference-gene selection, hierarchy construction, marker-panel selection, and model fitting. Holdout cells are used only for generating test pseudobulks, preventing test-cell expression from influencing assay design or training.

The flat assay is evaluated on targeted pseudobulk mixtures generated from discovery cells for training and from holdout cells for testing. The completed hierarchical experiments use the same discovery/holdout principle and independently generated hierarchical pseudobulk mixtures, with completed configurations covering two, three, and four top-level lineages and a pseudobulk-size comparison for the four-lineage hierarchy. The planned two-lineage, larger-pseudobulk interaction experiment is not treated as a completed configuration here.

Marker candidates, normalization references, hierarchy structure, and model parameters are derived from the discovery cells only. Reference genes are kept separate from marker candidates, and panel scores subtract reference-gene expression from marker-gene expression. Flat multi-gene panels and best-single-marker baselines are selected and evaluated on the targeted pseudobulk protocol. Hierarchical parent and conditional child panels are constructed from discovery-derived candidates for their respective lineage-level and within-lineage discrimination problems. All linear-regression predictions are clipped to [0, 1].

We evaluate two-, three-, and four-lineage agglomerative hierarchies constructed from discovery-derived cluster signatures. The completed sensitivity analysis includes the two-, three-, and four-lineage configurations at the initial hierarchical pseudobulk size, plus a four-lineage comparison at the larger completed pseudobulk size. In each hierarchical configuration, parent-lineage and conditional child models are fitted independently and their clipped predictions are multiplied to estimate the target fraction. Flat multi-gene panels and best-single-marker baselines are evaluated under the corresponding targeted pseudobulk protocols, using the same discovery/holdout design.

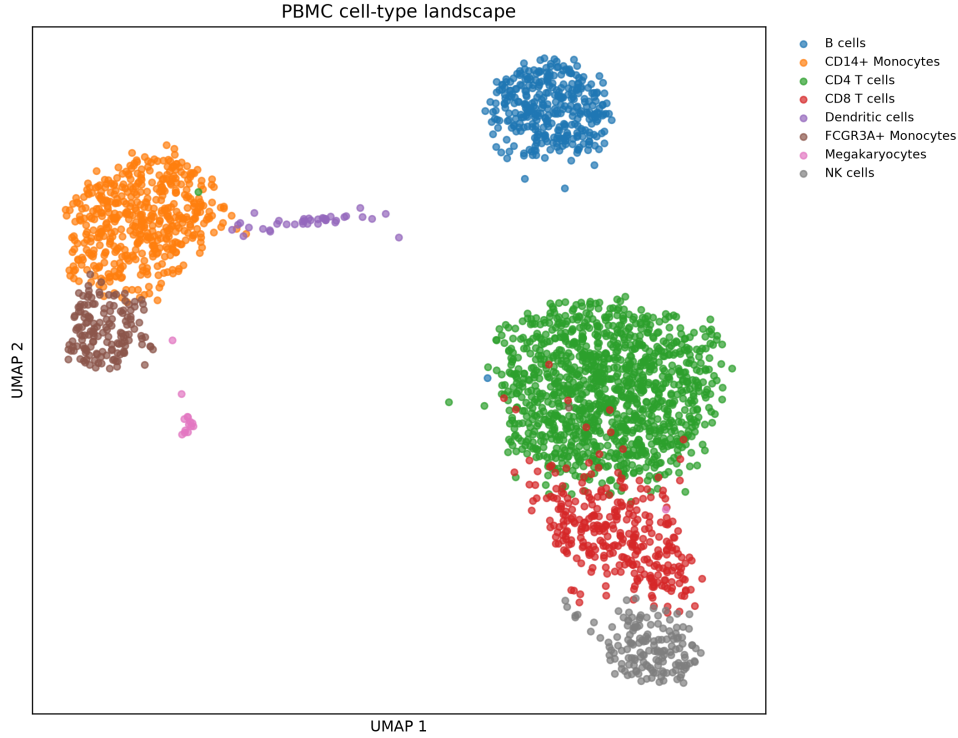


Figure 1: UMAP embedding of the supplied PBMC dataset, colored by the Leiden clusters used as cell-population labels.

A target is considered present for detection when its true fraction is at least $\tau = 0.35$. For flat assays, we report per-cluster and macro-averaged ROC AUC, Spearman correlation, Pearson correlation, MAE, RMSE, R^2 , and MAE within low- and high-fraction subsets. The hierarchical evaluation additionally reports rare-target average precision, sensitivity at a threshold determined by the 95th percentile of predictions on negative test mixtures, bias, parent-fraction MAE, conditional-fraction MAE, and product-propagation MAE. The latter compares the final prediction with the explicitly stored parent-child product and is an implementation consistency check, since the final prediction is defined from that product.

6 RESULTS

The supplied PBMC data provide a heterogeneous set of Leiden-defined cell populations with structure visible in the UMAP embedding (Figure 1). This organization motivates evaluating both broad lineage discrimination and confusion among related clusters. The hierarchy was constructed from discovery-cell signatures, so the displayed structure represents the expression geometry available during assay design rather than information from the held-out pseudobulks.

Across the completed hierarchical configurations, the flat multi-gene comparator remained quantitatively strong and relatively stable. Its macro ROC AUC ranged from 0.9816 to 0.9837, macro Spearman correlation ranged from 0.9440 to 0.9512, and macro MAE ranged from 0.0616 to 0.0640. In the two-lineage, 100-cell configuration, the flat panel achieved macro ROC AUC 0.9816, macro MAE 0.0640, and macro RMSE 0.0803, compared with 0.9823, 0.0658, and 0.0823 for the best single-marker baseline. Across completed configurations, the panel-versus-single-marker AUC difference was close to zero, ranging from -0.0007 to 0.0003 .

The per-cluster diagnostics show that aggregate flat-panel performance masks target-specific behavior. The ROC curves compare the selected panel with the best individual marker for each cluster, while the true-versus-predicted plots expose calibration across low, medium, and high target fractions (Figures 3 and 4). Residuals further show where errors vary with abundance rather than simply

Multi-gene panel versus best single-marker baseline — run_2

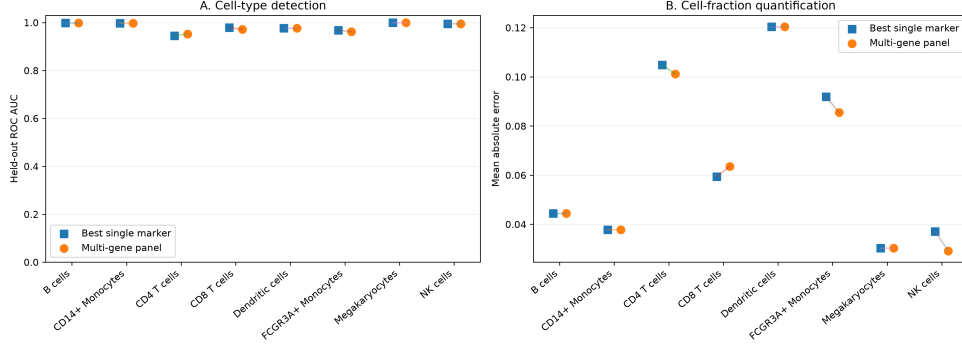


Figure 2: Per-cluster held-out ROC AUC and MAE for the flat multi-gene panel and best single-marker baseline in the two-lineage, 100-cell configuration.

Held-out cell-type detection performance — run_2

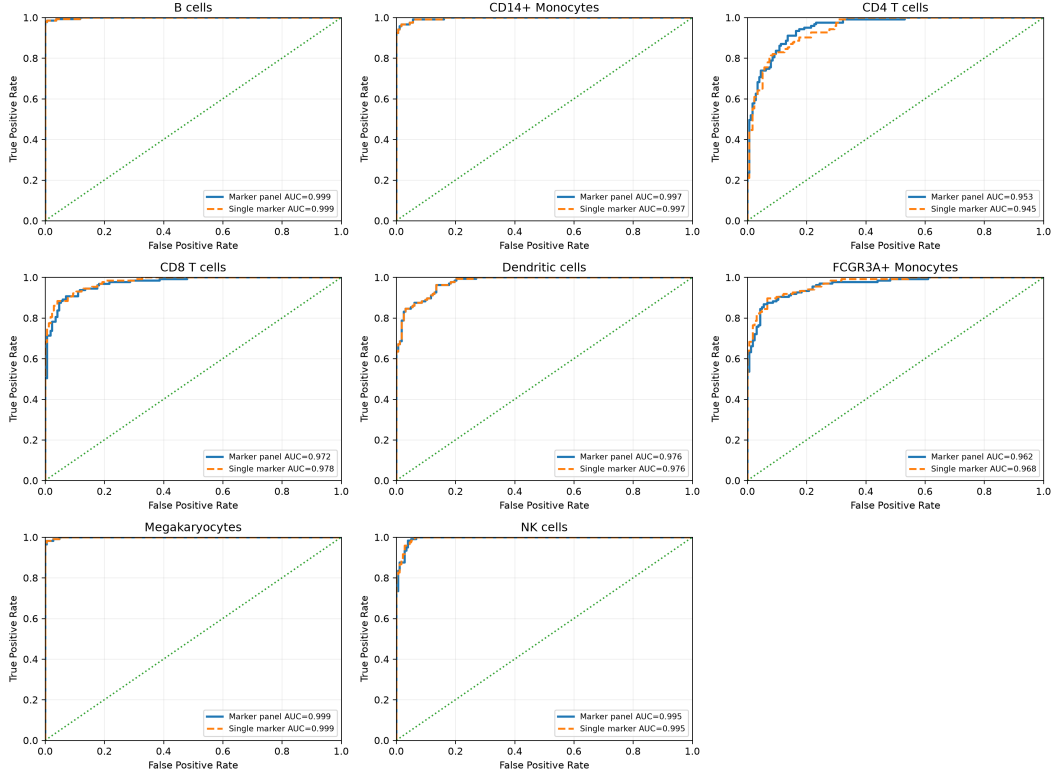


Figure 3: Per-cluster held-out ROC curves comparing flat multi-gene panels with best single markers in the two-lineage, 100-cell configuration.

reflecting random deviations around zero (Figure 5). These plots support using both detection and continuous-error metrics: strong ranking can coexist with abundance-dependent calibration error.

Hierarchical detection was sensitive to the number of top-level lineages. With 100-cell pseudobulks, the two-lineage configuration achieved ROC AUC 0.9745, rare-target average precision 0.9262, and fixed-specificity sensitivity 0.8253. The corresponding three-lineage configuration achieved 0.9573, 0.9042, and 0.8060, whereas the four-lineage configuration achieved 0.9304, 0.8171, and 0.6533. Thus, the coarsest tested hierarchy gave the strongest hierarchical detection metrics, but this result

Quantitative cell-fraction prediction — run_2

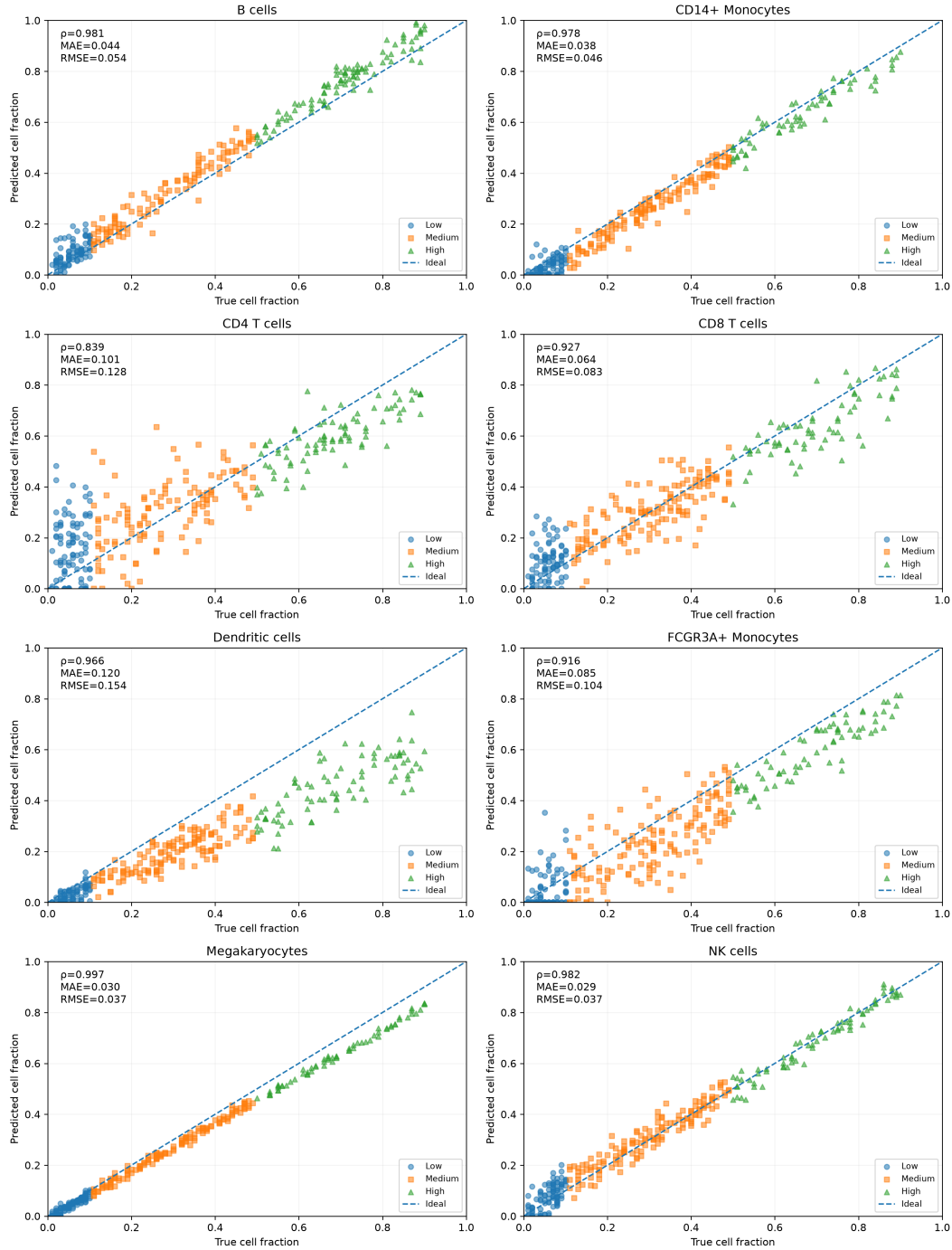


Figure 4: True versus predicted target fractions for the flat panels in the two-lineage, 100-cell configuration, stratified by abundance regime.

should not be interpreted as uniformly optimal because quantitative performance followed a different pattern.

Hierarchical quantitative estimation was worse than the matched flat panel in every completed configuration. For the two-lineage, 100-cell configuration, hierarchical MAE was 0.0996 versus

Prediction residuals across abundance regimes — run_2

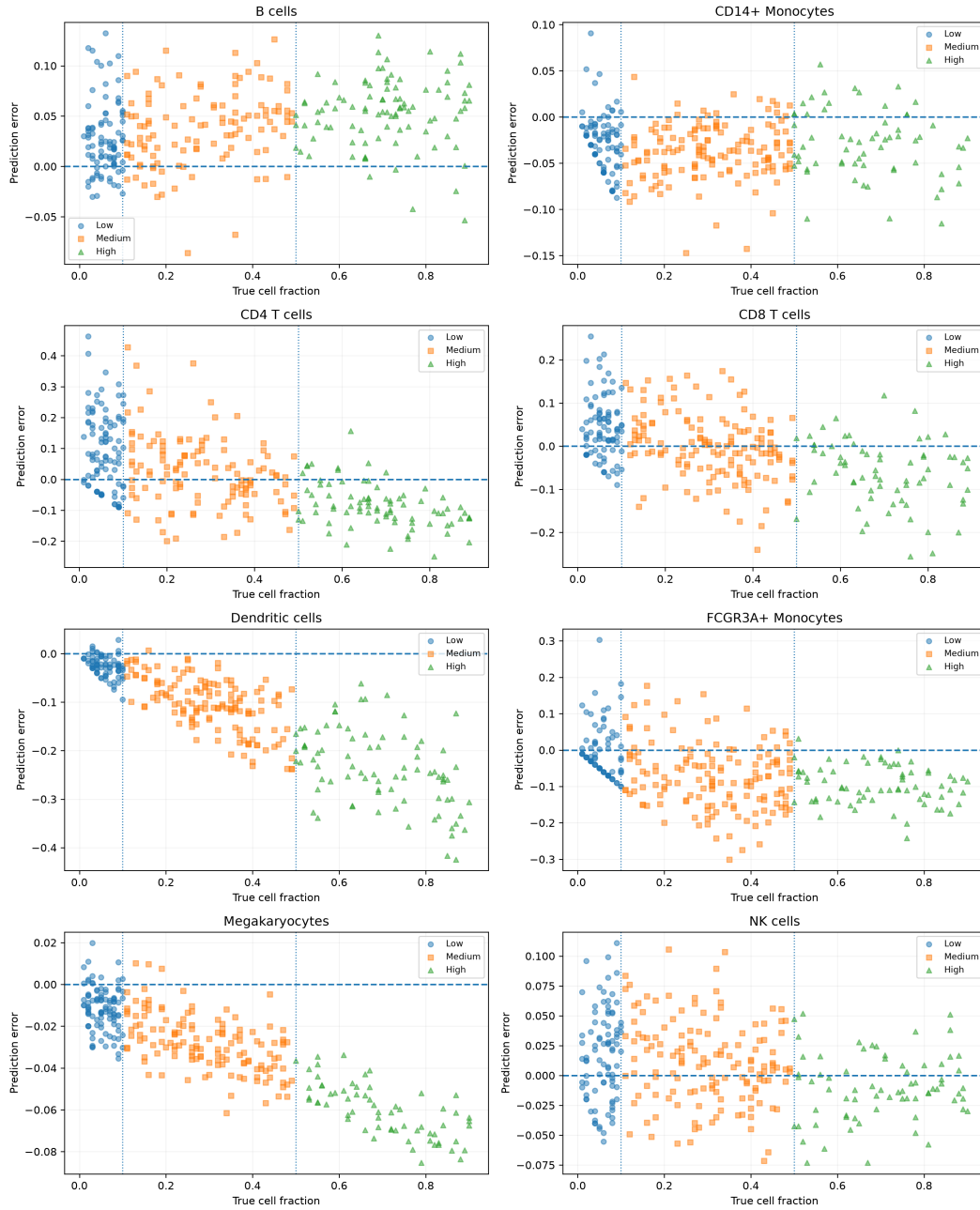


Figure 5: Flat-panel prediction residuals across target abundance regimes in the two-lineage, 100-cell configuration.

flat-panel MAE 0.0640, and hierarchical RMSE was 0.1184 versus 0.0803. The hierarchy also showed positive bias of 0.0777 and low-fraction MAE of 0.1166, compared with flat-panel low-fraction MAE 0.0477. The conditional child MAE was 0.1903, exceeding the parent-lineage MAE of 0.1400, indicating that sibling-level conditional estimation was a persistent source of error. In the three-lineage configuration, hierarchical MAE was 0.0862 and conditional MAE was 0.1738; in the four-lineage, 100-cell configuration, these values were 0.0857 and 0.1866, respectively.

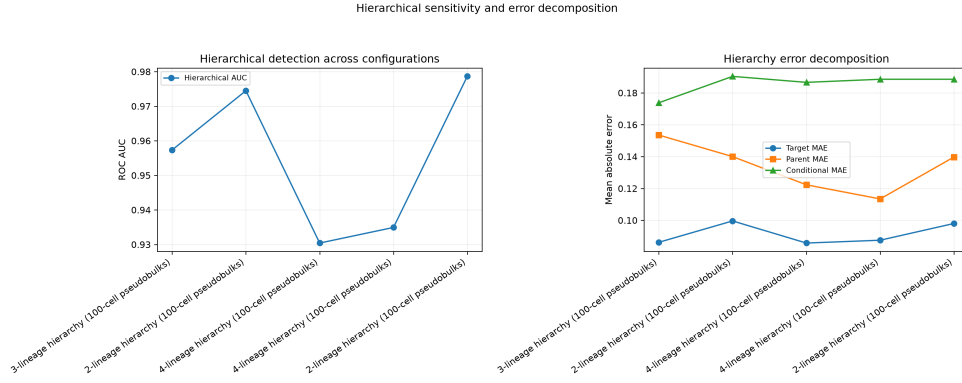


Figure 6: Hierarchical detection and error decomposition across completed hierarchy and pseudobulk configurations. Target, parent-lineage, and conditional child MAE are shown separately.

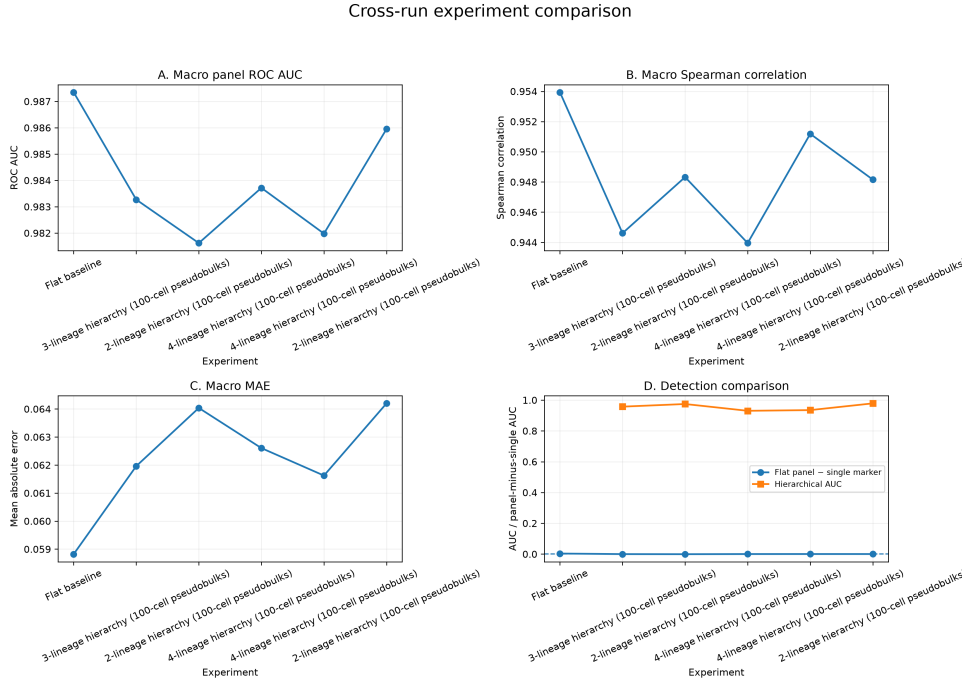


Figure 7: Comparison of completed configurations using flat-panel macro metrics and hierarchical detection metrics.

Increasing the four-lineage pseudobulk size from 100 to 250 cells produced mixed changes. Hierarchical ROC AUC increased slightly from 0.9304 to 0.9349, fixed-specificity sensitivity increased from 0.6533 to 0.6550, and parent-fraction MAE decreased from 0.1223 to 0.1134. However, hierarchical MAE increased from 0.0857 to 0.0875, Spearman correlation decreased from 0.8018 to 0.7902, and conditional fraction MAE increased from 0.1866 to 0.1885. The flat panel remained stable in the same comparison, with macro MAE c

The discovery-derived dendrograms and marker heatmaps show how the assay architecture changes with the lineage cut while retaining compact parent and child panels (Figures 8 and 9). The two-lineage structure separates broad groups before applying target-specific child panels. In contrast, the four-lineage configuration changes the parent partition without eliminating the conditional child error observed quantitatively. These structural plots therefore complement, rather than replace, the held-out performance comparisons.

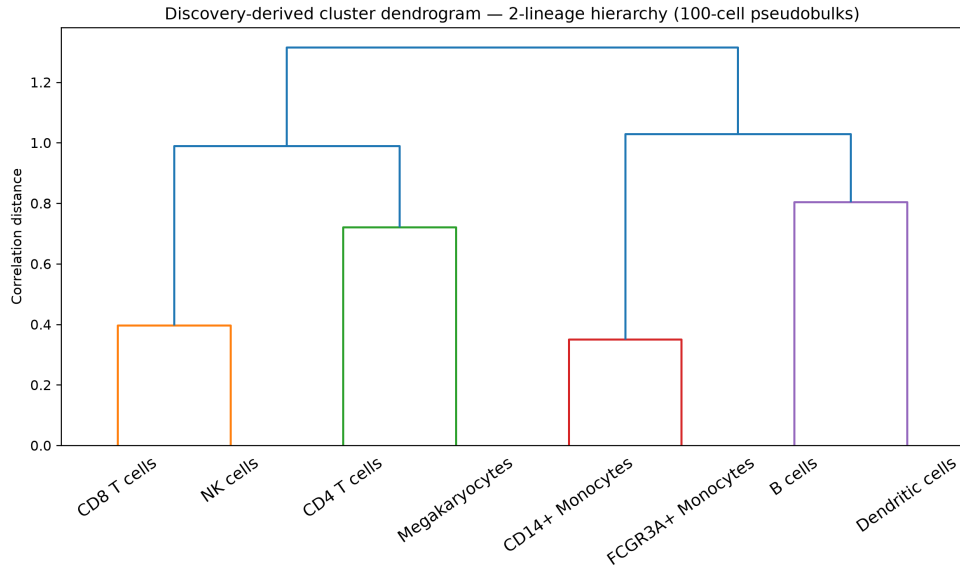


Figure 8: Discovery-derived cluster dendrogram for the completed two-lineage hierarchical configuration. Branch heights represent correlation distance among cluster signatures.

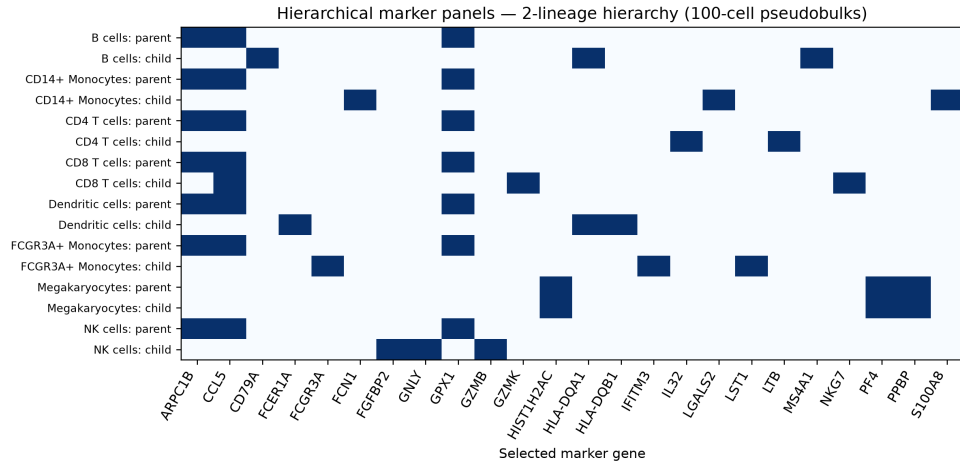


Figure 9: Parent and child marker membership for the completed two-lineage hierarchical configuration. Filled cells indicate selected genes in each target-specific panel.

Overall, the experiments provide a mixed result for coarse-to-fine assay design. Hierarchical organization can yield strong held-out detection, particularly for the two-lineage configurations, but the independently modeled parent and conditional fractions did not produce competitive quantitative estimates relative to flat panels. The reported product-propagation MAE is zero because the final prediction is defined as the parent prediction multiplied by the conditional prediction; it is therefore a construction check, not evidence that hierarchical error is absent. The completed evaluation supports a practical distinction between detection-oriented and quantification-oriented assay design: the hierarchical approach is promising for structured detection, whereas the flat panel is more reliable when accurate continuous fraction estimation is the primary objective.

7 CONCLUSIONS AND FUTURE WORK

We developed and evaluated a discovery-only hierarchical marker-panel framework for estimating cell-type fractions from bulk-like molecular measurements. The framework constructs a hierarchy from discovery-cell expression signatures, selects normalization references and parent/child marker panels using discovery data, and estimates each target fraction as the product of independently fitted parent-lineage and conditional child models. We evaluated two-, three-, and four-lineage configurations, varied hierarchical pseudobulk size, and compared the resulting assays with flat multi-gene panels and best-single-marker baselines on independent holdout-derived pseudobulks.

The experiments showed that hierarchical performance depends strongly on the lineage partition. Among the completed configurations, the two-lineage hierarchy with 100-cell pseudobulks produced the strongest hierarchical detection metrics, with ROC AUC 0.9745, rare-target average precision 0.9262, and fixed-specificity sensitivity 0.8253. However, its quantitative MAE was 0.0996, compared with 0.0640 for the matched flat panel. Across the hierarchy configurations, conditional child-fraction error remained high, ranging from 0.1738 to 0.1903, while increasing pseudobulk size did not consistently improve quantitative estimation. The flat multi-gene panel was comparatively stable, and its advantage over the best single marker was generally modest: in the completed four-lineage, 250-cell configuration, macro MAE was 0.0616 for the panel and 0.0639 for the single-marker baseline, while macro ROC AUC was 0.9820 and 0.9817, respectively.

Together, these results support a conditional use of hierarchical assays. A coarse-to-fine design can provide strong structured detection when broad lineage separation is favorable, but independently estimating and multiplying parent and conditional fractions can impair calibration and continuous fraction accuracy. The zero product-propagation error reported by the implementation is a consistency check arising from the definition of the final prediction, not evidence that hierarchical estimation has no error. Thus, when accurate quantitative fraction estimation is the primary objective in this setting, the simpler flat panel is better supported by the completed experiments; hierarchical panels are more appropriately viewed as a detection-oriented architecture whose quantitative advantages were not demonstrated in the completed evaluation.

Several extensions follow naturally from these findings. Future methods could fit parent and child components jointly under nonnegative or compositional constraints, calibrate conditional child models explicitly, and select hierarchy cuts using held-out quantitative objectives rather than detection alone. More demanding evaluations should include shifted-composition test mixtures, related-cluster false-positive analyses, bootstrap confidence intervals, and systematic sensitivity analyses over target abundance, sibling similarity, tree depth, and pseudobulk size. These developments would test whether hierarchical structure can retain its detection benefits while reducing the conditional-model error that limited quantitative performance in the present study.

This work was generated by THE AI SCIENTIST (Lu et al., 2024).

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